VINAY K R

Generative AI Engineer | LLM Engineer | Applied AI / RAG Engineer

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PROFESSIONAL SUMMARY

Generative AI and Software Engineer with nearly 3 years of experience building backend REST services, distributed application caching, and LLM-powered systems across fintech and gaming environments. Experienced in developing multi-agent LangGraph workflows, hybrid RAG pipelines, local inference gateways, semantic caching with Redis, and OpenTelemetry observability. Proven background in Node.js, TypeScript, Python, C#/.NET, and Angular with a strong focus on clean API design, system stability, and automated testing.

TECHNICAL SKILLS

- Generative AI & Agentic Systems: LangGraph StateGraph, supervisor and worker routing, state reduction, checkpointing, human-in-the-loop approval, RAG pipelines, relevance grading, query rewriting, context engineering
- Retrieval & Embeddings: dense vector and BM25 lexical retrieval, cosine similarity, reciprocal rank fusion (RRF), context reranking and compression, Redis 8 vector caching, local feature embeddings, document chunking
- LLM APIs & Frameworks: FastAPI, Pydantic v2, Python, REST APIs, Server-Sent Events (SSE), OpenAI-compatible API schemas, Model Context Protocol (MCP) local tool integration
- Observability & Evaluation: OpenTelemetry, OTLP tracing, Jaeger, custom span tracking, heuristic faithfulness and relevance scoring, regression testing, latency and error monitoring
- Languages & Core Stack: Python, TypeScript, Node.js, JavaScript, C#/.NET, SQL, Redis, PostgreSQL, SQLite, Angular, RxJS
- Engineering & Packaging: pytest, Ruff, Mypy, GitHub Actions CI/CD, Docker, Docker Compose, unit and integration testing, API schema validation

EXPERIENCE

Software Engineer — Full Stack & AI Applications | Liminal Custody (First Answer India Services Pvt Ltd) |
Nov 2025 – Mar 2026

- Implemented atomic bulk operations and transaction rollbacks in Node.js/TypeScript REST APIs across 5 core microservices, improving change safety and data consistency.
- Added Redis caching with ownership and expiry validation plus address risk integrations with short-circuit rule evaluation across 10,000+ daily checks, reducing external API latency and failure exposure.
- Implemented organization-scoped RBAC and step-up authentication after tracing multi-tenant authorization paths across Auth0, Angular guards/interceptors, and backend middleware for 100+ enterprise accounts.

Senior Associate Software Engineer | Light & Wonder (LNW India Solutions Pvt Ltd) | *Aug 2023 – Jul 2025*

- Diagnosed and fixed an RxJS/WebSocket subscription-lifecycle leak across 100+ continuously operating gaming device units, eliminating memory overhead and application crashes.
- Unified frontend crash reports and backend telemetry in a centralized error logging workflow across 3 engineering teams to accelerate bug triage and fault isolation.
- Developed audit-data capture, operational reporting, and Cypress end-to-end regression coverage across 50+ financial and player management workflows, strengthening release quality.

Full-Stack Intern | Light & Wonder (LNW India Solutions Pvt Ltd) | *Mar 2023 – Jul 2023*

- Completed a 16-week C#/.NET Core and Angular internship and delivered a production game-recommendation application with user feedback and ratings.

PROJECTS

FastAPI GenAI Agent Patterns | Python, FastAPI, LangGraph, Redis 8, OpenTelemetry
https://github.com/vi-nayKR/fastapi-genai-agent-patterns

- Built a provider-neutral FastAPI service with a typed LangGraph supervisor that plans and routes research, coding, and compliance specialists, bounds iterations, interrupts high-risk tasks for human approval, and streams token events via SSE.
- Implemented two-stage Redis 8 caching combining exact SHA-256 keys with native vector set search and local feature embeddings, reducing redundant model lookups to sub-5ms response times.
- Instrumented API, agent, approval, and cache paths with OpenTelemetry tracing exported to Jaeger; validated stability with automated GitHub Actions CI tests.

Enterprise Agentic RAG Platform | Python, LangGraph, FastAPI, Hybrid Retrieval, RRF
https://github.com/vi-nayKR/enterprise-agentic-rag-platform

- Developed an agentic RAG system with document chunking, LangGraph routing between retrieval and tool branches, relevance grading, automated query rewriting, and citation-formatted answers.
- Implemented hybrid retrieval by running cosine vector search and BM25 lexical search concurrently, fusing result rankings with Reciprocal Rank Fusion (RRF) and applying context reranking.
- Integrated local tool server prototypes over JSON-RPC protocols for database and API queries; validated response grounding using automated quality metrics.

Local LLM Inference Gateway | Python, FastAPI, OpenAI-Compatible API, SSE
https://github.com/vi-nayKR/local-llm-inference-gateway

- Created an OpenAI-compatible FastAPI gateway supporting model list and chat completion endpoints with streaming SSE token delivery and local response fallbacks.
- Built a process-local semantic query cache with feature hashing vectors, statistics tracking, prompt injection detection, and PII sanitization utilities.
- Designed workflow configurations for instruction dataset formatting, QLoRA fine-tuning metadata, and Ollama model packaging.

Multimodal Document Intelligence | Python, Pydantic, FastAPI, Structured Extraction
https://github.com/vi-nayKR/multimodal-document-intelligence

- Modeled page layout bounding boxes and Pydantic invoice schemas for structured document extraction and validation.
- Implemented table reconstruction with Markdown/CSV export, line-item arithmetic validation, token grounding checks, and FastAPI review routes.

LLM Observability & Evaluation Platform | Python, FastAPI, Pydantic, Evaluation
https://github.com/vi-nayKR/llm-observability-eval-platform

- Built a custom LLM observability platform tracking hierarchical Pydantic trace records, execution timing contexts, token counts, cost estimates, and error rates.
- Implemented automated evaluation metrics for answer faithfulness, relevance, context precision, and toxicity with regression reporting dashboards.

EDUCATION & RESEARCH

B.E. Computer Science & Engineering, Siddaganga Institute of Technology | 2019–2023 | CGPA 8.65/10
Co-author: *“Data Visualisation of Time Tradable Assets Using Machine Learning,” IEEE, 2023*